

Overfitting

Project: Responsible Plant Classifier

1. Overfitting

Teach the learning cycle

Primary target: Compare training and validation evidence for model capacity

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Case exposure check: this source example appears in ai-2/16, ai-4/27, ai-4/30. Reuse a familiar case for

recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Reproducible Experiments

Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Record enough detail to reproduce a run

Reference example, not the only acceptable wording: Teacher explanation: Mean 70 percent; range 80-60=20 percentage points, across the three declared runs.

Train, Validation, Test

Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

Look for: Select a model using a declared validation rule

Reference example, not the only acceptable wording: Teacher explanation: Choose A: validation MAE 3 ties C, and A wins the simplicity tie. B has validation MAE 5 despite training MAE 0.

Absolute Error and MAE

Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate MAE MSE and RMSE

Reference example, not the only acceptable wording: Teacher explanation: $(1+2+3)/3=2$ cm.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A has training 98%, validation 58%; B has training 86%, validation 81%. Compute gaps and select a candidate with a qualification.

Concept to explain

An overfit model captures training-specific noise, producing a large gap between training and unseen performance.

Term · Meaning

Term: overfitting

Meaning: fit to noise rather than general pattern

Term: complexity

Meaning: model capacity to form detailed boundaries

Term: generalization gap

Meaning: train score minus unseen score

Term: regularization

Meaning: constraint discouraging excessive complexity

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on overfitting and model capacity. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer. Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

No; it may fit peculiarities of the training sample.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank evidence table

Saved classroom project records for critique; no personal student dataset required

End goal for the pictorial case

By the end, I can explain overfitting using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics;

schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Compare simple and complex models using train and validation evidence.

MODEL IDEA An overfit model captures training-specific noise, producing a large gap between training and unseen performance.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Simple tree: train 82%, validation 78%; gap 4 points. Deep tree: train 100%, validation 65%; gap 35 points.

Draw a fresh data → model → prediction → evaluation trace:

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

STEP 1 Simple tree: train 82%, validation 78%; gap 4 points.

STEP 2 Deep tree: train 100%, validation 65%; gap 35 points.

STEP 3 Deep tree's perfect training result is weaker unseen evidence.

STEP 4 Prune, regularize, improve representative data, or features—then rerun fixed validation cases.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

Concrete teaching case

A scores 100% train, 60% validation; B scores 85%, 80% on the same sets.

Compute gaps and select by validation accuracy.

Exact explanation to model

A gap 40 percentage points; B gap 5. B has better shared validation accuracy. A suggests overfitting, but investigate distribution differences too.

Explain the added worked example

A more flexible model can represent complicated relationships and accidental details. Capacity can help when there is enough relevant evidence, but perfect training performance is not the goal. Compare training and validation using the same metric while checking leakage and representativeness.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data

split visible as you reason. Earlier weeks supplied the model-building process; this week examines overfitting and model capacity.

On the same regression split, a constant has training MAE 5 and validation 6, a line has training 2 and validation 3, and a highly flexible curve has training 0 and validation 8. Under smallest validation MAE, choose the line. The curve has the lowest training loss but the worst validation result.

Before continuing, underline the decision rule and circle the evidence that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim. Keep the distinction between an observed result and a proposed next step.

Key terms

Term · Meaning

Term: Capacity

Meaning: Flexibility of a model to represent patterns

Term: Overfitting

Meaning: Learning details that fail to generalize well

Term: Underfitting

Meaning: Failing to capture useful structure

Term: Generalization

Meaning: Performance on relevant unseen examples

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Overfitting and model capacity at a glance

Part · Evidence or action · Interpretation

Part: Constant

Evidence or action: Training 5

Interpretation: Validation 6

Part: Line

Evidence or action: Training 2

Interpretation: Validation 3

Part: Flexible curve

Evidence or action: Training 0

Interpretation: Validation 8

The curve result is consistent with overfitting, but a gap alone does not prove its cause. A changed population, bad labels or a split error could also produce a gap. Inspect the procedure before claiming a diagnosis. The constant may miss useful structure, but these numbers alone do not identify the true relationship.

Does a training-validation gap alone prove overfitting? Explain the steps or evidence behind your response.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger

claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Overfitting

See overfitting as a gap requiring investigation, not a label based only on model size.

The facts we will use

Toy model Small: training 85%, validation 80%. Large: training 100%, validation 60%. Same split and scoring rule.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/27

Training and validation accuracy

percent



Bar length shows the supplied value; the marked baseline is zero.

Overfitting

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute gaps

Small gap=5 percentage points; Large gap=40 percentage points.



2. Select development candidate

Small has the higher validation accuracy despite its lower training fit.



3. Investigate large model

Check capacity, training choices and data mismatch without tuning on final cases.

CHANGE ONE THING

Large's validation accuracy rises to 82% after a development revision.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Compute gaps

What happens and why: Small gap=5 percentage points; Large gap=40 percentage points.

Step: 2. Select development candidate

What happens and why: Small has the higher validation accuracy despite its lower training fit.

Step: 3. Investigate large model

What happens and why: Check capacity, training choices and data mismatch without tuning on final cases.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Check capacity, training choices and data mismatch without tuning on final cases. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

A gap can also reflect data shift or sampling noise. Inspect evidence before attributing every gap to capacity.

Change one thing in the pictured case

Change: Large's validation accuracy rises to 82% after a development revision.

Prediction: Its gap is 18 percentage points; it now beats Small on this validation set.

Reason: Validation performance improved, but the training-validation gap is not a complete release rule.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: A scores 100% train, 60% validation; B scores 85%, 80% on the same sets.

Write the question to explain: Compute gaps and select by validation accuracy.

Reveal after predictions: A gap 40 percentage points; B gap 5. B has better shared validation accuracy. A suggests overfitting, but investigate distribution differences too.

Keep for an independent attempt: Model A has training accuracy 99% and validation accuracy 70%; B has 88% and 84%. Compute each gap and discuss the evidence.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: No. It is consistent with overfitting but may also reflect shift, data quality or split problems. Exit response: Check data provenance, split grouping, label quality and repeated development results; inspect representative residuals without using final test feedback to tune.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: Can a zero training error guarantee good validation performance? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify capacity in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: select from evidence; interpret a gap. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between capacity and overfitting. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Describe one fair next development comparison.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Compare training and validation evidence for model capacity; Explain overfitting without assuming every gap proves it. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion task: Does a training-validation gap alone prove overfitting?

Teacher check: Teacher explanation: No. It is consistent with overfitting but may also reflect shift, data quality or split problems.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Overfitting and model capacity. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Choose the line with validation MAE 3. The curve training MAE 0 does not compensate for validation MAE 8.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Does 100% training accuracy prove Large was always best? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

No. Originally Large scored 60% on validation versus Small 80%; training perfection alone did not establish generalisation. Changed-case check: Its gap is 18 percentage points; it now beats Small on this validation set. Validation performance improved, but the training-validation gap is not a complete release rule.

Changed-case reasoning: Its gap is 18 percentage points; it now beats Small on this validation set. Validation performance improved, but the training-validation gap is not a complete release rule.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected evidence before running.
2. Complexity Trial: fit paper trees of increasing depth, plot train/validation results, and identify the first clear overfit pattern.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR The model with the highest training accuracy is the best model.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Choose the line with validation MAE 3. The curve training MAE 0 does not compensate for validation MAE 8.

A stronger claim makes a stronger project: Check data provenance, split grouping, label quality and repeated development results; inspect representative residuals without using final test feedback to tune.

The new case can be copied from the example: The independent task changes the evidence. Require a fresh explanation: B wins the validation tie by the declared simplicity rule. C is not selected merely for its zero training error.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

A gap 40 percentage points; B gap 5. B has better shared validation accuracy. A suggests overfitting, but investigate distribution differences too.

Independent question

Model A has training accuracy 99% and validation accuracy 70%; B has 88% and 84%. Compute each gap and discuss the evidence.

Expected independent answer

A's gap is 29 percentage points; B's is 4. A shows stronger evidence of overfitting, while B performs better on this validation set.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Higher training accuracy always identifies the better generalizing model.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

A gap=40 percentage points; B gap=5. B is the stronger candidate on shared validation accuracy. A suggests overfitting, but distribution differences or leakage checks remain relevant. Use untouched final

data after selection.

Explanations for the additional workbook examples

W1 Select from evidence

Choose among the three worked candidates and explain the training trap.

Teacher explanation: Choose the line with validation MAE 3. The curve training MAE 0 does not compensate for validation MAE 8.

W2 Interpret a gap

Does a training-validation gap alone prove overfitting?

Teacher explanation: No. It is consistent with overfitting but may also reflect shift, data quality or split problems.

W3 Propose a controlled change

Describe one fair next development comparison.

Teacher explanation: Compare a simpler curve against the current curve on the same validation rows, retaining the same metric and training-only fitting; log the change.

W4 Apply independently

Models A training 4 validation 5, B training 1 validation 2, C training 0 validation 2. Lowest validation MAE then simpler; B is simpler than C. Choose and justify.

Teacher explanation: B wins the validation tie by the declared simplicity rule. C is not selected merely for its zero training error.

W5 Limit a diagnosis

What extra evidence would you inspect before concluding a model memorized noise?

Teacher explanation: Check data provenance, split grouping, label quality and repeated development results; inspect representative residuals without using final test feedback to tune.

Assess independent transfer

Selected case: Model A training 96%, validation 62%; model B training 88%, validation 79%. Compare gaps and choose a development candidate, assuming the same valid evaluation conditions.

Selected case answer

A gap 34 percentage points; B gap 9. B has better validation performance and a smaller gap, so is the development candidate under the supplied conditions. Training accuracy alone is insufficient. Freeze decisions before a separate final evaluation.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Responsible Plant Classifier

Check Plant Classifier train-validation gaps and limit complexity before final test release.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 28: Learning Curves. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank evidence table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use capacity and overfitting in full sentences. Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: B wins the validation tie by the declared simplicity rule. C is not selected merely for its zero training error. Fresh retry training/validation D 3/4 E 0/7 F 2/3: choose F under lowest validation loss. Explain why E is not automatically best. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Learning curves and variability. Carry forward the saved evidence and the distinction between development choices and justified claims.

Plan the delayed return

Suggested return points in this level: ai-4/28, ai-4/30, ai-4/33. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.